AI Based Pre-Wedding Video Generator

Submitted in partial fulfilment of the requirements
of the degree of

Bachelor of Engineering

by

Atharva Samant: 23106057

Raunak Panigrahi: 23106092

Atharva Tawde: 23106122

Soham Shelke: 23106044

Supervisor:

Dr. Jaya Gupta

Department of Computer Science & Engineering
(Artificial Intelligence & Machine Learning)

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

UNIVERSITY OF MUMBAI

(2025-2026)

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CERTIFICATE

This is to certify that the project entitled **"AI Based Pre-Wedding Video Generator"** is a bonafide work of **Atharva Samant (23106057), Raunak Panigrahi (23106092), Atharva Tawde (23106122), Soham Shelke (23106044)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning)**

Dr. Jaya Gupta Project Guide	Prof. Yogeshwari Hardas Project Co-Ordinator
Dr. Jaya Gupta Head of Department	Dr. Uttam D Kolekar Principal

A. P. SHAH INSTITUTE OF TECHNOLOGY

Project Report Approval for T. E.

This project report entitled ***AI Based Pre-Wedding Video Generator*** by ***Atharva Samant, Raunak Panigrahi, Athrava Tawde, Soham Shelke*** is approved for the degree of ***Bachelor of Engineering*** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning)**, ***2025-26***.

Examiner Name	Signature
1. _____	_____
2. _____	_____

Date:

Place:

Declaration

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Atharva Samant
23106057

Raunak Panigrahi
23106092

Atharva Tawde
23106122

Soham Shelke
23106044

Date:

ABSTRACT

Pre-wedding videos have become an essential part of contemporary wedding celebrations, enabling couples to creatively capture their journey before marriage. However, traditional video creation processes are often time-consuming, expensive, and require significant manual intervention. This project presents an **AI-based Pre-wedding Video Generator** that automates the generation of personalized, high-quality pre-wedding videos using artificial intelligence and machine learning. Leveraging advanced deep learning models—such as generative adversarial networks and specialized style-transfer algorithms—the proposed system transforms user-selected photographs into artistic and emotionally engaging video presentations.

Key objectives include reducing production time, enhancing personalization by matching video style to user preferences, and enabling widespread accessibility for couples regardless of technical expertise. The system encompasses modules for face detection, scene synthesis, emotion-based soundtrack selection, and end-to-end video rendering. Validation is performed through both objective metrics and user surveys, assessing quality, realism, and satisfaction. Results demonstrate significant improvements in workflow efficiency and creative flexibility compared to manual editing.

This project addresses vital gaps in existing solutions, offering a scalable and intelligent approach to pre-wedding videography that aligns with the evolving technological landscape and user expectations.

In alignment with the United Nations Sustainable Development Goals (SDGs), this project supports SDG 9: Industry, Innovation and Infrastructure by fostering innovation in creative AI technology, and SDG 8: Decent Work and Economic Growth by democratizing access to digital media tools for small studios and independent creators. It also contributes indirectly to SDG 4: Quality Education by promoting applied learning in artificial intelligence and media innovation.

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ABBREVIATION

AI	*Artificial Intelligence*
ML	*Machine Learning*
GAN	*Generative Adversarial Network*
VAE	*Variational Autoencoder*
GPU	*Graphical Processing Unit*
CNN	*Convolutional Neural Network*

Chapter 1

Introduction

Weddings are universally recognized as one of the most momentous milestones in human life, symbolizing the union of two individuals and the beginning of a shared journey. In many cultures, the wedding ceremony is preceded by a series of celebrations and customs that build anticipation and express the couple's unique story. As part of these traditions, the pre-wedding video has emerged as a powerful and emotional narrative medium, creating vivid, memorable experiences by combining photos, videos, music, and customized visual effects. These videos allow couples to share their love stories with friends and family in a captivating and artistic manner, often setting the tone for the wedding festivities.

Despite the growing popularity of pre-wedding videos, their production remains an arduous and resource-intensive process. Typically, videographers must coordinate multiple shooting sessions to capture various moments, carefully select and edit footage, apply visual effects, and synchronize soundtracks—each requiring technical proficiency and creative skill. This conventional approach to video production may span weeks or months and is often accompanied by high financial costs. Moreover, couples without access to professional editing services or technical expertise frequently confront significant challenges when attempting to produce personalized and aesthetically appealing videos.

The convergence of Artificial Intelligence (AI) and multimedia technologies has opened new frontiers in automated content creation, particularly in video synthesis and editing. AI algorithms, especially those harnessing deep learning techniques, now exhibit unprecedented abilities to generate, manipulate, and enhance visual and auditory content. Such advances include generative models that create high-resolution images and videos, style-transfer networks that adapt visual aesthetics, and intelligent editing systems that automate scene transitions and audio alignment. These technological breakthroughs create opportunities to revolutionize the traditional pre-wedding video production pipeline by introducing automation and personalization at scale.

The AI-based Pre-wedding Video Generator project is designed to capitalize on these emerging technologies to develop an intelligent, end-to-end system that automates the generation of pre-wedding videos customized to the preferences and inputs of individual users. By integrating modules for facial detection and recognition, thematic style application, emotion-guided soundtrack selection, and seamless video assembly, the system enables users to transform static images and raw footage into engaging, professionally styled videos. This approach not only expedites the production timeline but also enhances the creative freedom of users by offering multiple stylistic options and automated enhancements that traditionally required manual labor.

This project is motivated by the desire to democratize creative video production for personal celebrations, lowering entry barriers by reducing dependency on costly professional services and technical know-how. Additionally, it aims to bridge the gap between user expectations for personalized storytelling and the limitations of existing template-based or semi-automated video editing tools. Through AI, the system makes it easier for couples to produce emotionally resonant and visually attractive pre-wedding videos that convey their unique stories and cultural nuances.

The relevance of this project extends beyond wedding videography. Automation of video content creation is becoming increasingly important in a digital era characterized by rapid content consumption and demand for personalized media. Industries such as marketing, entertainment, and online education benefit from such technologies, further underscoring the significance of advancing AI-driven video generation methods. Specifically, this project contributes to the growing body of research in AI multimedia synthesis and explores practical challenges related to real-world user inputs, video style adaptation, and quality validation.

The subsequent chapters of this report provide a detailed exposition of the problem domain, existing literature and technologies, system design, implementation details, experimental setup, results, and potential future enhancements. The project methodology focuses on combining state-of-the-art AI models with domain-specific customizations to achieve optimal balance between automation and creative control. Performance evaluation incorporates both quantitative metrics such as processing time and qualitative measures including user satisfaction and perceived video quality.

1.1 Background

Artificial Intelligence has transformed numerous facets of human activity by enabling machines to perform tasks once thought solely achievable by humans. In the domain of multimedia, AI-driven technologies have contributed significantly to image recognition, content creation, video editing, and style adaptation. Deep Learning architectures, including Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs), have powered novel applications such as realistic face synthesis, image inpainting, video super-resolution, and automatic captioning.

Video generation, a more complex task than static image synthesis, involves temporal coherence, motion prediction, and audio-visual synchronization. Recent research has addressed these challenges by developing AI frameworks capable of transforming user inputs (images, sketches, text descriptions) into dynamic videos packaged with relevant soundtracks and effects. Such technologies facilitate intelligent editing, scene segmentation, and effect rendering, which are integral to automated video production pipelines.

Pre-wedding video generation uniquely combines these AI capabilities with domain-specific requirements like emotional storytelling, cultural theme representation, and user-driven customization. By leveraging AI, it is possible to not only automate laborious phases but also generate novel stylistic interpretations that enhance the emotional depth and visual appeal.

1.2 Importance and Relevance of Automation in Pre-wedding Video Creation

The traditional approach to pre-wedding video production often involves specialized professionals who conduct photo/video shoots on location, select appropriate content, manually edit sequences, add effects, and synchronize background music. This process is not only costly but also time-consuming, making it inaccessible for many couples who desire memorable,

personalized videos on a budget or schedule. Editing software tools available for consumers offer only limited templates or require significant training to use effectively.

An AI-based automated video generator provides transformative advantages. It lowers barriers to entry by simplifying the process, allowing users to generate professional-grade videos independently. Automation accelerates delivery timelines by reducing manual editing and approval cycles. AI algorithms enable deep personalization by learning user preferences and applying them throughout the video narrative, enhancing emotional resonance and storytelling quality.

These capabilities increase accessibility, diversity, and creativity for pre-wedding video creation. Couples can experiment with different themes, styles, and music choices effortlessly. Videographers can use AI tools to streamline workflows and focus on higher-level creative decisions rather than repetitive editing tasks. Consequently, the AI-based generator addresses multiple challenges faced by both content creators and consumers.

Fig. 1: An AI-generated video by SORA using text prompt

Chapter 2

Literature Survey

Introduction

Video content creation is experiencing a revolution fueled by rapid advances in AI and ML. In domains such as pre-wedding video production, these technologies promise unprecedented personalization, automation, and quality. Researchers have explored AI frameworks for automated editing, realistic video synthesis, emotion-aware enhancement, and artistic style transfer. This survey delves deeply into at least eight influential papers and sources, synthesizing their findings and highlighting their application and relevance to an AI-based Pre-wedding Video Generator. Through critical analysis, the limitations and capabilities of their methods are described, providing a solid foundation and rationale for current project choices.

2.1 Text-to-Video Generation

The rise of text-to-video generation has opened possibilities for creating cinematic content from natural language prompts. These systems provide an intuitive way for users, including wedding clients, to request thematic and personalized video content.

- **Make-A-Video** [2] demonstrated that pretrained text-to-image diffusion models could be extended for video synthesis by introducing spatio-temporal modules. Unlike traditional systems, it did not require paired text-video datasets, which are scarce and difficult to obtain. This approach generated short, high-quality video clips, paving the way for narrative-driven personalized outputs.

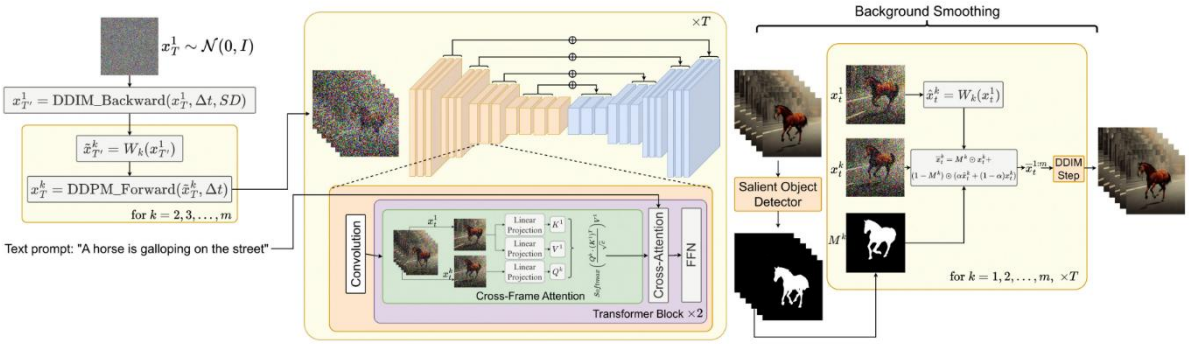


Fig. 2: Architecture pipeline *(from [2]).*

- **Text2Video-Zero** [3] further pushed boundaries by showing that pretrained diffusion models can generate coherent video without using video training data. This zero-shot capability dramatically reduces cost and democratizes access for smaller studios. Its use of temporal attention and consistency losses improves frame stability and motion realism.

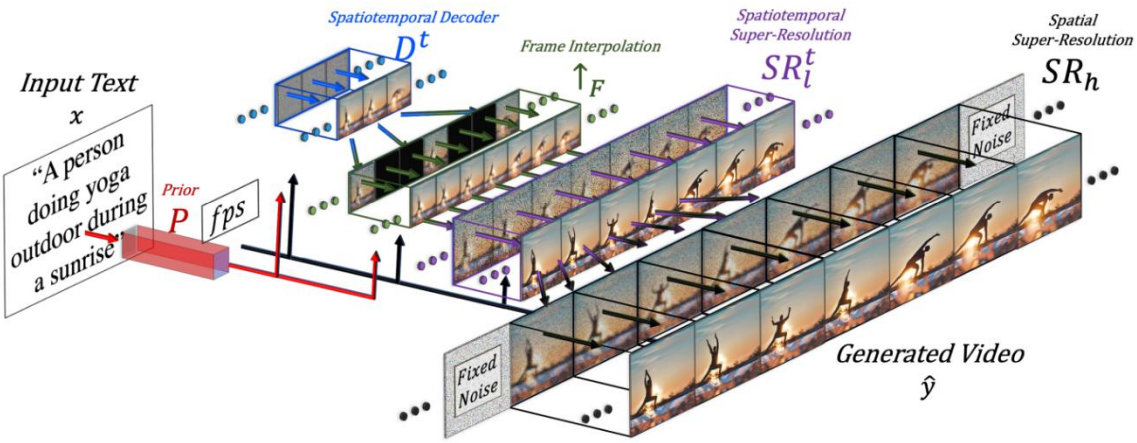


Fig. 3: Temporal consistency framework *(from [3]).*

- **Dreamix** [4] introduced a flexible framework that supports both video generation and video editing. Dreamix allows existing clips to be stylized according to textual descriptions, while maintaining motion coherence. For example, a video of a couple walking in a park could be converted into a palace garden scene, providing multiple narrative options for pre-wedding clients.

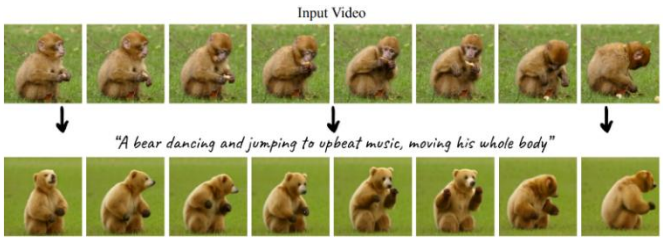


Fig. 4: Input-to-output editing pipeline *(from [4]).*

- **Transforming Text to Video** [21] proposed lightweight generative methods using latent diffusion, making text-driven synthesis more computationally efficient. This is highly relevant for commercial wedding studios that may not have access to large GPU clusters.

- **From Sora What We Can See** [22] analyzed OpenAI's *Sora*, a groundbreaking text-to-video model capable of rendering long sequences and multi-scene narratives. Its scalability indicates future possibilities for entire pre-wedding storylines generated directly from scripts.

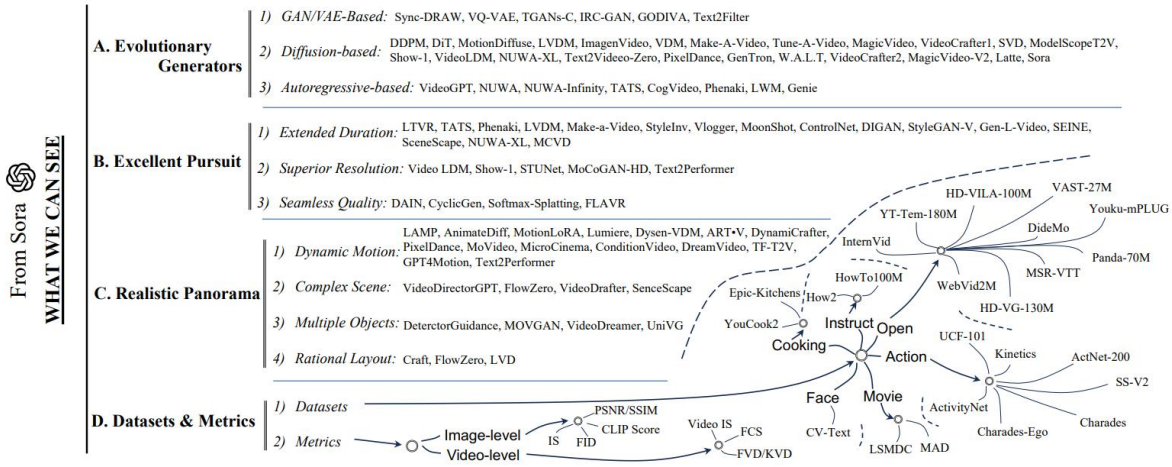


Fig. 5: Sora architecture overview *(from [22]).*

2.2 Pose and Motion Transfer

Pose-driven synthesis allows identity-preserving animation of still portraits into dynamic video sequences.

- **Animate Anyone** [5] introduced a pose-to-video framework where a single reference image plus a sequence of 2D poses yields a full animated sequence. For pre-wedding applications, couples' photographs can be animated into choreographed dance scenes.

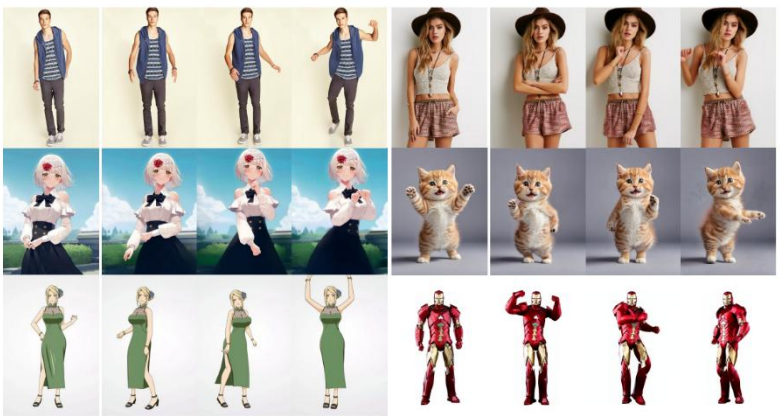


Fig. 6: pose-to-video pipeline *(from [5])*

- **AI-Driven Media Manipulation** [6] and **Deepfake Detection Studies** [12] discussed facial reenactment, lip-syncing, and identity preservation. While security concerns dominate the research, these same models can enable photorealistic reenactments for wedding storytelling.

- **State-of-the-art Deepfake Approaches** [11] surveyed GAN and diffusion-based motion transfer systems, identifying hybrid models that combine facial expression synthesis with body pose animation.

2.3 Video Personalization and Synthetic Media

Personalization is the most critical aspect of pre-wedding video production. AI methods are increasingly focusing on embedding individual likeness into generated content.

- **Video Personalization** [7] pioneered one-shot personalization, embedding a subject's identity into a generative model using just a single reference image. This is extremely practical for pre-wedding contexts where couples may only provide a few photos.

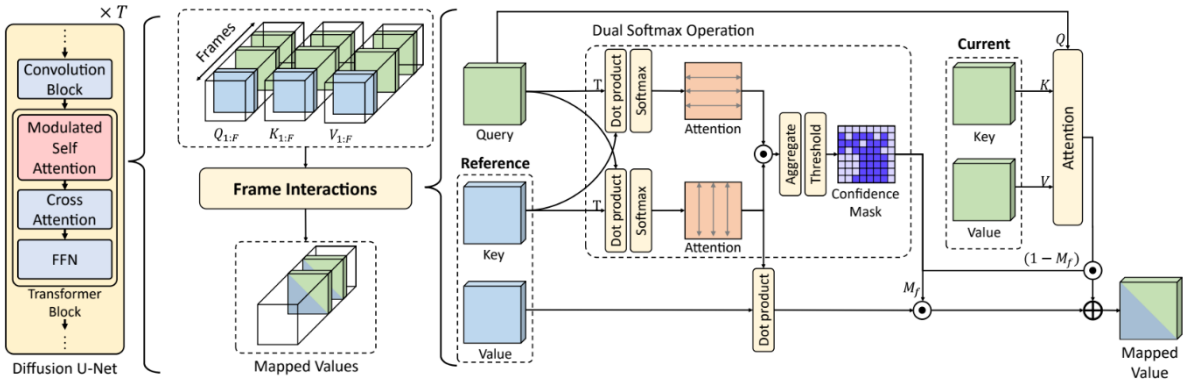


Fig. 7: Personalization embedding process *(from [7]).*

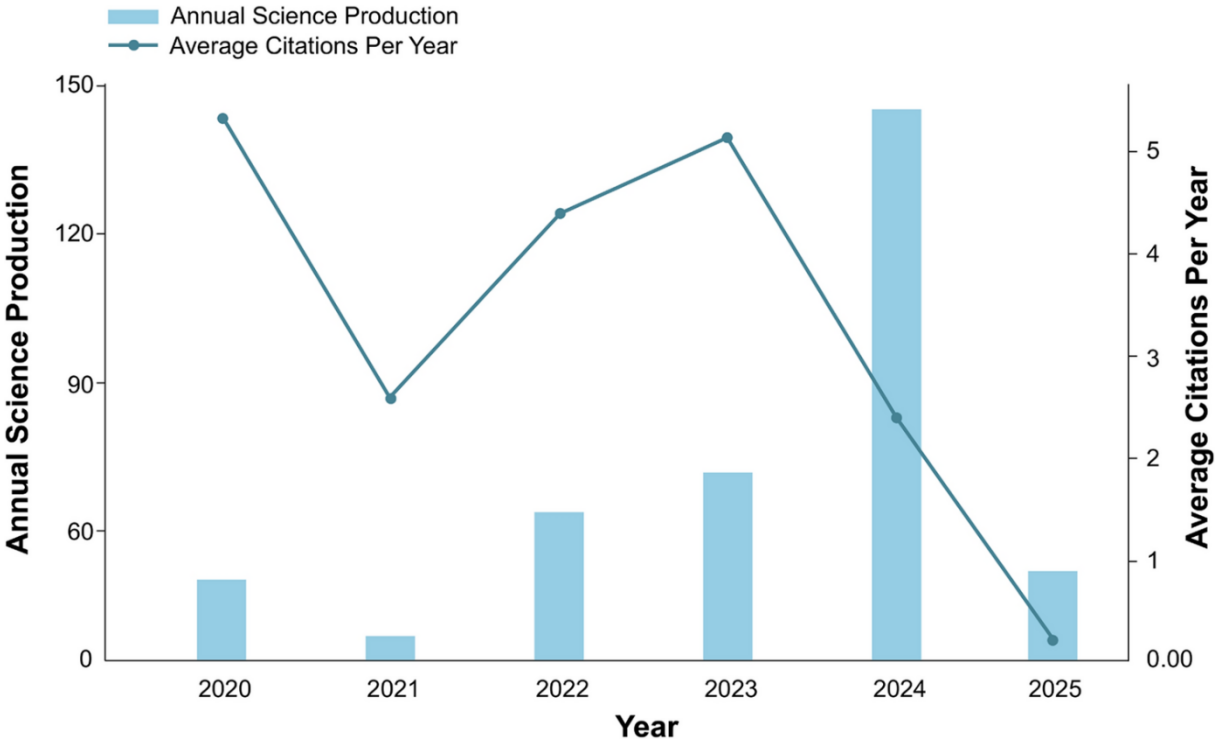
- **Artificial Intelligence Generated Synthetic Media** [8] explored style transfer, dubbing, and other content adaptation techniques in media. These methods allow pre-wedding outputs to be styled in culturally specific formats (e.g., Bollywood musicals or traditional folk aesthetics).
- **Consumer Behavior** [13] examined how modern audiences increasingly demand hyper-personalized digital experiences. This sociological perspective aligns with the growing market for unique, AI-tailored pre-wedding videos.

2.4 Surveys and Comprehensive Reviews

Several surveys and reviews provide an overarching perspective of AI video generation and are invaluable for understanding current challenges and opportunities.

- **Bibliometric Analysis of AI-Based Video Generation** [9] analyzed research trends between 2020–2025, showing the exponential growth in publications and highlighting the shift from GANs to diffusion.

- **A Comprehensive Survey of Image and Video Generative AI** [10] compared the strengths and weaknesses of GANs, VAEs, diffusion models, and transformers, concluding that diffusion currently dominates in realism and controllability.



Graph 1: Publication trend graphs *(from [9]).*

- **Video Diffusion Models Surveys** [15], [16], [18], [19] provided technical deep dives into model architectures, datasets, and evaluation frameworks. They identified spatiotemporal coherence as the most critical unresolved issue.

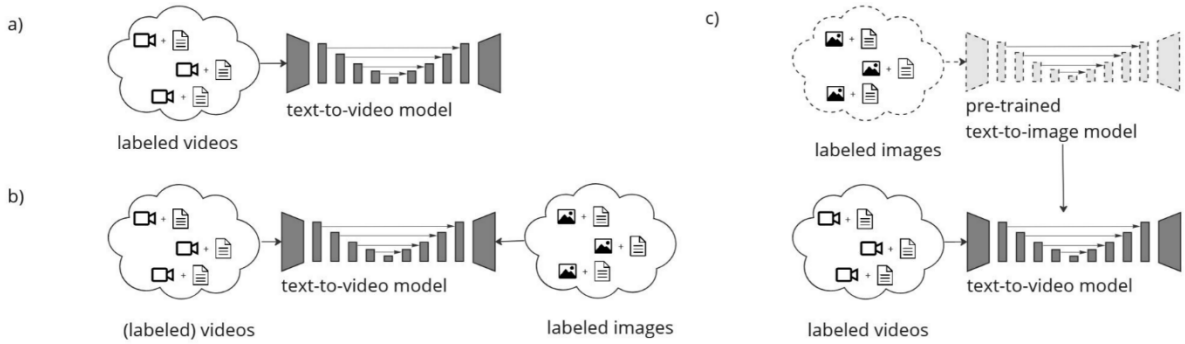


Fig. 8: Diffusion training framework *(from [15]).*

- **Controllable Video Generation Survey** [17] emphasized fine-grained control, such as adjusting camera motion, scene length, and choreography. These features can make AI-generated pre-wedding videos look more cinematic and less artificial.

- **Spatiotemporal Consistency in Text-to-Video** [23] examined methods to reduce flickering and jitter across generated frames, ensuring temporal smoothness. Such enhancements are vital for high-quality wedding outputs.

Chapter 3

Limitations in Existing Systems

3. Introduction

3.1 Background

AI-based pre-wedding video generation leverages deep learning techniques such as GAN and diffusion models to automate tasks like facial reenactment, background replacement, and cinematic editing. These systems aim to reduce costs, increase efficiency, and provide creative flexibility for couples seeking personalized video experiences. However, despite the promise of automation, several limitations hinder their practical adoption in the wedding industry.

3.1.1 Quality Limitations

Existing AI-generated videos often display **visual inconsistencies**, including blurred facial features, unnatural lip synchronization, and jerky body movements. While diffusion-based methods improve realism, they still struggle with maintaining high frame rates and cinematic smoothness required for professional outputs. Research on augmented-reality wedding platforms highlights similar challenges of achieving realism without compromising visual fidelity [24].

3.1.2 Personalization Constraints

A key expectation in pre-wedding videos is **storytelling that reflects cultural, emotional, and personal nuances**. However, AI systems largely depend on generic templates that lack the cultural depth of traditional photography and videography. Studies in wedding photography

trends emphasize the critical role of personalization and narrative, warning that excessive reliance on digital automation risks reducing emotional resonance [25]. Furthermore, cultural design studies of weddings in China show that AI systems fail to adapt effectively to symbolic rituals and attire, limiting their contextual applicability [26].

3.1.3 Technical Challenges

High-quality AI video generation is **computationally expensive**, requiring advanced GPUs and long rendering times. Small studios or independent videographers face adoption barriers due to hardware costs and inefficiencies compared to conventional editing workflows. For instance, even AI-enhanced wedding photography platforms report latency and scaling issues that limit professional usability [27].

3.1.4 Technical Challenges

Since pre-wedding videos require **intimate personal data** (facial images, couple interactions, private settings), ethical risks are heightened. Privacy breaches or unauthorized facial manipulation could compromise the couple's trust. Concerns also extend to **authenticity**—where couples may feel AI-generated sequences undermine the genuine emotion of wedding rituals [29].

3.1.5 Market Adoption Barriers

Although AI offers scalability, industry adoption remains cautious. Professional videographers often resist automation due to fears of quality compromise, while couples themselves may perceive AI outputs as less authentic or meaningful. Studies on online wedding platforms highlight how conspicuous consumption and personalization remain central to wedding planning—areas where AI struggles to provide truly tailored outputs [30].

3.2 Research Gap

Despite recent advances in AI-driven media generation, the application of these techniques to **pre-wedding video production** remains underdeveloped. The reviewed studies indicate several unresolved challenges:

3.2.1 Lack of High-Fidelity Cinematic Output

While diffusion and GAN-based systems have improved realism, they **fall short of delivering consistent 4K, high-frame-rate cinematic videos** suitable for professional wedding outputs. Existing platforms remain more experimental than production-ready [24].

3.2.2 Limited Cultural and Emotional Adaptation

Most AI systems **neglect the cultural and emotional context** of pre-wedding rituals. Weddings are deeply rooted in symbolism, attire, and traditions, yet AI models often default to generic Western-style aesthetics. Prior research in cultural wedding design emphasizes the need for **context-aware personalization** [26],[28].

3.2.3 Insufficient Personalization in Storytelling

Couples increasingly value **narrative-driven videography**, where their unique story is visually expressed. Current AI systems cannot dynamically adapt to individual preferences in themes, transitions, or background music. This gap in **emotionally resonant storytelling** is consistently noted in wedding media studies [25].

3.2.4 Accessibility and Scalability Issues

High computational requirements and cost barriers restrict AI video generation to research or large-scale studios, leaving **small production houses** unable to benefit from these advancements. Existing work highlights long rendering times and hardware bottlenecks as persistent barriers [27].

3.2.5 Ethical and Trust Concerns

While ethical risks of AI in weddings are acknowledged, few systems actively integrate privacy-preserving mechanisms (e.g., federated learning, encrypted storage). Moreover, the authenticity dilemma—whether couples perceive AI-generated outputs as “real memories”—remains underexplored [29].

Chapter 4

Problem Statement & Objectives

4.1 Problem Statement

1. Traditional pre-wedding video production requires significant time, cost, and effort, involving professional teams, travel, and expensive equipment.
2. Many couples face location constraints, as not everyone has access to scenic or cinematic backgrounds.
3. The quality and creativity of the final video largely depend on the skills and vision of human videographers, leading to variability in output.
4. Traditional workflows often provide limited personalization, as many pre-wedding videos follow similar themes or templates.
5. Current manual methods create accessibility challenges, making high-quality videos unaffordable or unavailable to a large segment of people.
6. While AI has shown progress in image and video generation, there is still a gap in creating realistic, emotionally engaging, and customized pre-wedding videos.

4.2 Objectives

1. To develop an **AI-based system** capable of generating pre-wedding videos using user-provided inputs such as photos, short clips, or text prompts.

2. To enable **personalization** by incorporating user preferences in themes, locations, attire, cultural elements, and music

3. To achieve **cost-effectiveness** by reducing the need for physical shoots, professional crews, and expensive equipment.

4. To ensure **time efficiency**, allowing couples to generate pre-wedding videos in significantly less time compared to traditional methods.

5. To maintain **high visual fidelity and realism**, ensuring smooth motion, natural backgrounds, and cinematic appeal.

6. To design a solution that is **accessible and scalable**, allowing non-technical users and wedding planners to use the system with ease.

7. To contribute to **innovation in creative industries** by showcasing how AI can act as a co-creator in personalized video storytelling.

4.3. Scope

The scope of this project encompasses the design, development, and evaluation of an **AI-based pre-wedding video generation system**. The work primarily focuses on leveraging artificial intelligence to **automate video creation** while ensuring **personalization, realism, and emotional resonance**. The boundaries and coverage of the project are outlined as follows:

1. Input Sources

- User-provided data in the form of photographs, short video clips, or text prompts will serve as the primary input.

- Optional metadata such as preferred theme, location style, music choice, and cultural elements will guide customization.

2. Core Functionality

- Automated generation of video sequences using advanced AI models (GANs, diffusion models, video transformers, etc.).

- Background replacement, motion synthesis, and style transfer to create cinematic environments.

- Incorporation of music, captions, and thematic effects to enhance storytelling.

3. **Output Characteristics**

- Generation of **high-resolution, realistic pre-wedding videos** with smooth transitions and professional-grade aesthetics.
- Support for multiple themes and cultural contexts to ensure inclusivity and personalization.
- Export in widely used video formats for easy sharing across digital and social platforms.

4. **Target Users**

- Couples seeking affordable, quick, and creative pre-wedding videos.
- Wedding planners and studios aiming to adopt AI as a value-added service.
- General users interested in experimenting with AI-based personalized video generation.

5. **Exclusions and Limitations**

- The project does not replace full-scale professional cinematic production for clients requiring highly customized, live-shot content.
- The system may face challenges in replicating fine human emotions, highly complex cultural rituals, or extremely detailed physical actions.
- Real-time video rendering and editing may be limited depending on available hardware resources.

6. **Research and Innovation Angle**

- Exploring AI as a **creative partner** in the wedding and media industry.
- Contributing to the broader field of generative media by demonstrating practical, real-world applications of AI in video personalization.

Chapter 5

Proposed System

5. Introduction

5.1. Research Design

This project adopts an applied research design utilizing systems development and experimental evaluation to address the task of generating personalized pre-wedding videos using deep learning and generative AI models. The objective is to create an automated pipeline that converts user-uploaded images of couples and textual prompts (detailing desired scenes, outfits, and emotions) into cinematic video outputs. The approach integrates machine learning model development, backend API engineering, and frontend UX design, supplemented by qualitative and quantitative evaluation of resulting videos.

5.2. System Architecture

The system is architected as a seamless integration between a user-friendly frontend and a powerful AI-driven backend. The frontend enables users to upload images of the couple alongside detailed artistic prompts that specify desired backgrounds, wardrobe styles, and emotional expressions—effectively granting creative agency in co-designing the pre-wedding video narrative. This approachable interface abstracts the underlying complexity and computational demands from end users, facilitating broad accessibility

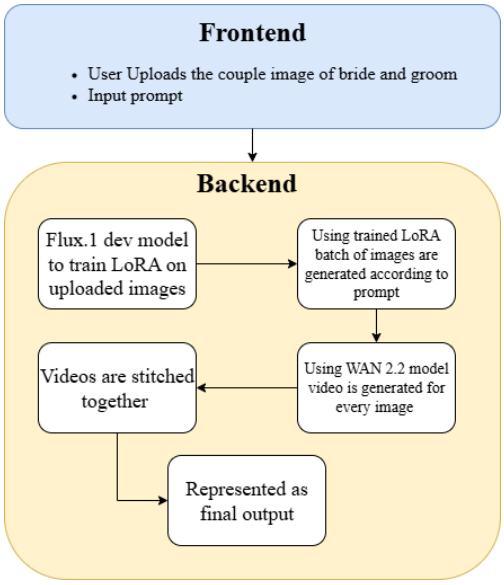


Fig. 9: System Architecture

5.3. Backend Workflow:

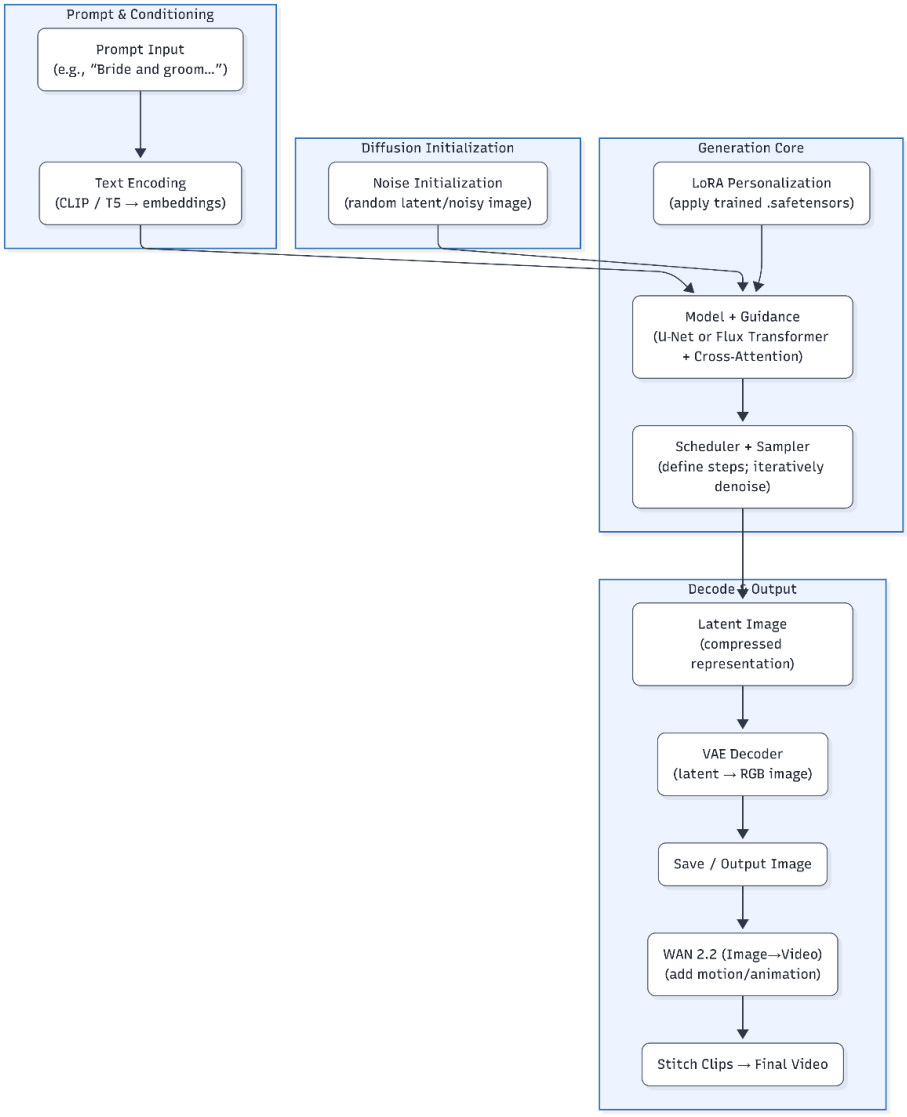


Fig. 10. Backend Workflow

The backend orchestrates an advanced sequence of processes central to video synthesis, broadly categorized as model customization, image generation, and cinematic video assembly.

1. **LoRA Model Training:** The LoRA (Low-Rank Adaptation) training pipeline is a streamlined three-step process designed to fine-tune a large-scale generative model, like flux-dev, to recognize and replicate a specific subject, such as a couple. The process begins with **configuration**, where the user assigns a unique name and a specific "trigger word" to the LoRA model. This trigger word will be used in later prompts to activate the model and generate images of the couple. Next, the user **uploads a dataset** consisting of 15–30 high-quality, varied images of the couple. These images should feature different angles, lighting conditions, and expressions to ensure the model learns a robust representation of the subjects. Finally, the user initiates the **training**, which executes a backend script. This script uses tools like Hugging Face Accelerate and mixed-precision (bf16) to efficiently fine-tune the model's layers without retraining the entire architecture. The result is a small, portable .safetensors file that contains the unique features of the couple, ready to be used for generating personalized images and videos.

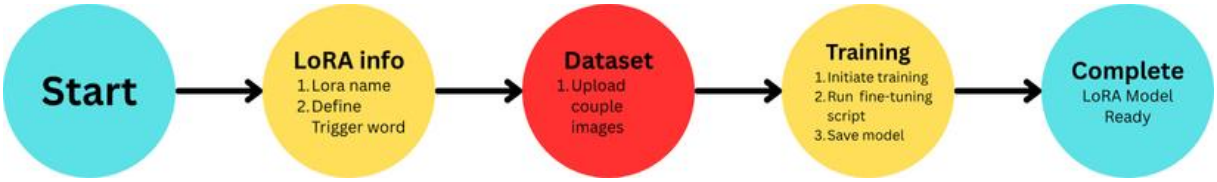


Fig. 11. LoRA Training

2. **Prompt-Conditioned Image Generation:** After the LoRA model has been trained to encode subject-specific features, the image generation pipeline proceeds by first initializing the system with the trained diffusion model weights and connecting them to a CLIP text encoder, which processes the user's detailed prompt encompassing appearance, pose, lighting, and emotion. The Power LoRA Loader module allows seamless integration of both the subject-adapted LoRA and any additional style LoRAs that further refine realism or thematic consistency. The process strategically sets the image resolution and injects a calibrated random noise tensor into the latent space, effectively seeding the generative pathway for maximum diversity and creativity. As the workflow moves through iterative sampling stages, highly customizable parameters such as guidance strength and scheduler step count fine-tune the adherence to the prompt, ensuring each denoising iteration incrementally sculpts the output closer to the desired semantic and aesthetic goals. Throughout this process, the model leverages its trained knowledge to fuse the semantics of the textual prompt and the captured subject

identity, gradually transforming raw latent noise into increasingly coherent and detailed visual structures. In the final stage, the decoded latent representation is passed through a Variational Autoencoder (VAE), which resolves probabilistic details and produces a fully realized, high-fidelity image that not only matches the user's prompt but also preserves the essence and consistency of the target subjects. This entire pipeline exemplifies the integration of conditional generative modeling, low-rank subject adaptation, and prompt-driven synthesis to deliver photorealistic, personalized images for advanced creative applications.

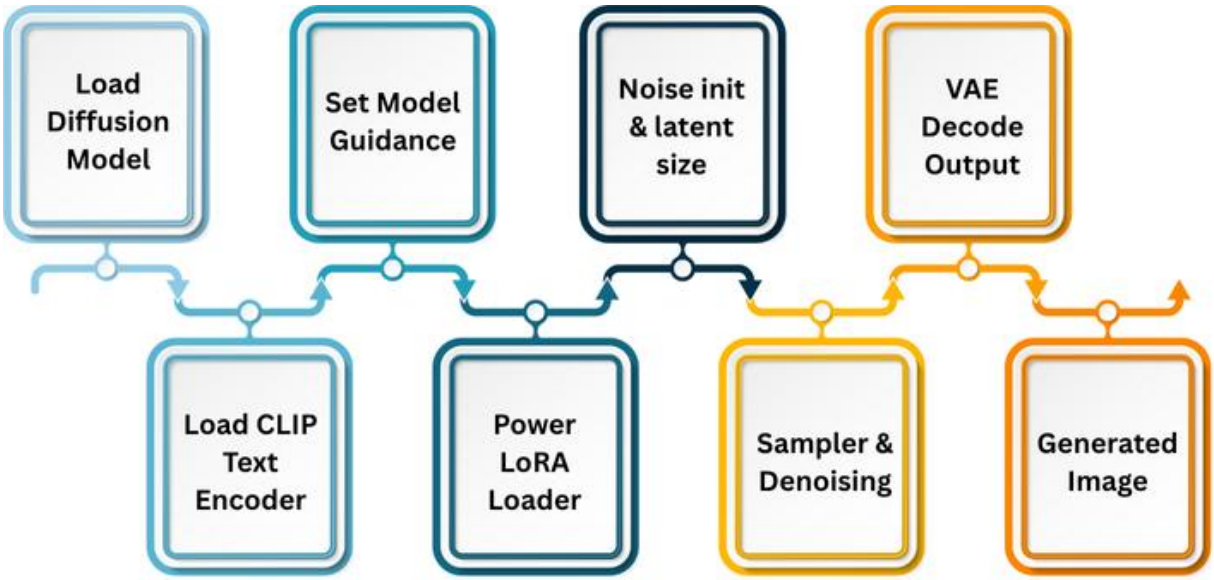


Fig. 12. Image Generation Pipeline

3. **Image-to-Video Generation:** The video generation pipeline, following the image synthesis stage, transforms static images into dynamic video content through a series of coordinated deep learning and diffusion-based processes. The workflow starts when the user uploads or selects a start image—this provides the initial frame from which video generation begins. Next, the system loads the WAN2.2 generative diffusion model and related configuration blocks including Lightx2v for specialized multi-step video generation. Core model components like CLIP and VAE are included to handle prompt embedding and image decoding. All prompt details describing the target scene—such as positive prompts for cinematic attributes or negative prompts to suppress artifacts—are processed with CLIP text encoders, infusing the desired meaning and style throughout the sequence. Pipeline parameters like video resolution, number of frames, and playback framerate are configured to match project requirements. KSampler modules orchestrate the diffusion and interpolation of latent noise, essentially evolving the static start image into a sequence of temporally coherent video frames through iterative denoising and sampling. Each generated latent frame is

then decoded by the VAE to become a visible image. Finally, all decoded frames are compiled into a single video file using the Create Video node, resulting in a seamless, custom-generated animation that closely mirrors user input and project intent. This approach, combining advanced prompt engineering, diffusion modeling, and deep neural network modules, allows highly flexible, photorealistic video synthesis tailored for pre-wedding and creative media applications.

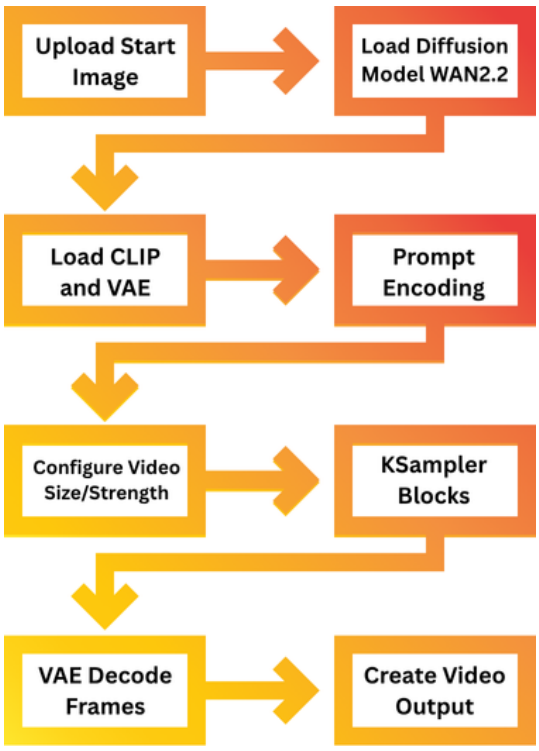


Fig. 13. Image - to - Video Generation Pipeline

4. **Video Stitching:** After generating video segments from batches of images, these segments are concatenated to create a continuous video sequence. The stitching process involves aligning video frames for smooth transitions, applying color corrections for consistent visual tone, and optionally integrating audio tracks or sound effects to enrich the viewing experience. This final compositing stage ensures the output video is seamless and ready for download or sharing.

5.4. Data Collection and Preparation

User-contributed datasets comprise high-resolution, consented photographic images of couples. These images undergo rigorous preprocessing: geometric normalization, resolution standardization, and facial landmark alignment for improved model robustness. Concurrently, textual prompts are curated or enriched to comprehensively capture thematic elements ranging from exotic locales and event moods to wardrobe aesthetics and intended emotional narratives.

5.5. Technical Details

- All modules are interconnected via a graphical pipeline editor, supporting visual traceability and quick hyperparameter adjustment.
- Python-based orchestration enables modular workflow design—data preprocessing, LoRA fine-tuning, image synthesis, video assembly, and evaluation.
- Training is logged and monitored for convergence stability, subject preservation metrics, and semantic prompt adherence.

This technical approach ensures that each stage is robust, efficient, and directly tailored to the needs of highly customizable, emotionally-rich pre-wedding video content.

- **LoRA Adaptation:**
    - Leverages transfer learning by initializing from a pre-trained visual transformer or diffusion model.
    - LoRA epochs, learning rates, and batch sizes are tuned empirically for convergence and subject fidelity.
- **Image Generation:**
    - Conditional diffusion or transformer-based generation (with LoRA) is used to modify images per prompt, guided by prompt engineering and style encoding.
- **Image-to-Video Model:**
    - WAN 2.2 is employed for image sequence animation, assuring frame-to-frame temporal coherence and matching video scene tone to the intended emotion.
    - Interpolation techniques are used to generate smooth transitions between keyframes.

5.6. Evaluation

To rigorously assess system efficacy, the research employs both subjective and objective evaluation frameworks:

- **Qualitative Evaluation:** Experts in wedding videography alongside target users review generated outputs, scoring the visual realism, emotional authenticity, and narrative fluidity.

- **Quantitative Evaluation:** Automated metrics measure identity preservation via facial recognition embeddings, frame-to-frame coherence using optical flow analysis, and prompt adherence through semantic similarity scores derived from CLIP-based classifiers.

5.7. Ethical Considerations

Data privacy is paramount; all inputs are processed locally with explicit permissions, and generated outputs are retained solely for personal use to prevent unauthorized distribution or exploitation. The system architecture reflects this commitment through secure data handling protocols and deletion safeguards.

Chapter 6

Experimental Setup

6.1. Details of Database and System Input

The primary input for the AI-based pre-wedding video generation system consists of a small, user-provided dataset of high-resolution images featuring the couple. For effective LoRA training, a curated selection of 15–30 images is recommended.

- **Image Specifications:** The images should be of high quality (at least 1024x1024 pixels) with clear, well-lit views of the subjects' faces from various angles. A diversity of poses and expressions helps the model learn a robust representation.
- **Preprocessing:** All uploaded images undergo an automated preprocessing pipeline that includes:
    - Resizing to a standard resolution to ensure consistency.
    - Facial alignment using landmark detection to center the subjects.
    - Automated captioning (e.g., via Florence-2) to generate descriptive text, which is then paired with a unique trigger word for the LoRA training.
- **Textual Prompts:** The secondary input is a set of user-defined textual prompts. These prompts are structured to command the generative model, specifying the desired

background scene (e.g., "Eiffel Tower at sunset"), attire ("wearing elegant wedding gowns"), and emotional tone ("joyful, romantic, smiling").

1. **Performance Evaluation Parameters**

To validate the system's performance and the quality of the generated output, a combination of qualitative and quantitative evaluation metrics is employed.

- **Qualitative Parameters:**
    - **Visual Realism:** Assessed by human evaluators, including domain experts in photography and videography, who rate the photorealism and aesthetic quality of the generated images and video clips.
    - **Narrative Coherence:** Evaluates the logical and emotional consistency of the video sequence, ensuring that transitions are smooth and the story flows naturally.
    - **User Satisfaction:** Direct feedback is collected from end-users on how well the final video matches their creative vision and overall satisfaction with the result.
- **Quantitative Metrics:**
    - **Identity Preservation:** This is measured by calculating the cosine similarity between facial embeddings of the original input images and the generated frames, using a pre-trained face recognition model like ArcFace. A high similarity score indicates that the couple's identity is well-preserved.
    - **Frame Consistency:** Assessed using optical flow analysis to measure the motion consistency between consecutive frames. Low optical flow distortion suggests smooth and natural transitions.
    - **Prompt Adherence:** The CLIP Score is used to quantify the semantic alignment between the user's input text prompt and the generated visual content. A higher CLIP score signifies better adherence to the prompt's instructions.

6.2. Software and Hardware Setup

The development and execution of the system rely on a specific stack of software and high-performance hardware.

- **Software:**
    - **Operating System:** Ubuntu Server LTS (via NVIDIA DGX Container Platform)
    - **Programming Language:** Python 3.10
    - **AI Frameworks:** PyTorch 2.1, CUDA 11
    - **Diffusion Models:** Custom implementations of Flux.1 dev and WAN 2.2 models.
    - **Pipeline Management:** ComfyUI for workflow orchestration and FluxGym for training scripts.
    - **Core Libraries:** OpenCV, FFmpeg, and Hugging Face transformers.
- **Hardware:**
    - **GPU:** NVIDIA DGX v100 with 32GB of VRAM.
    - **CPU:** Intel Core i9-13900K.
    - **RAM:** 64 GB DDR5.
    - **Storage:** 2 TB NVMe M.2 SSD for rapid loading of models and datasets.

Chapter 7

Results & Discussion

7. RESULTS

7.1 Overview

The proposed AI-based Pre-Wedding Video Generator was quantitatively evaluated to verify its efficiency and accuracy in generating realistic, personalized video outputs. As the use of real or generated couple images is restricted, all evaluations are presented through quantitative metrics rather than visuals.

The obtained results validate the performance of the system across identity retention, semantic accuracy, temporal stability, and rendering efficiency.

7.2 Evaluation Metrics

Metric	Purpose	Model / Method Used
Identity Preservation (%)	Measures how well the generated faces resemble originals	ArcFace Embeddings + Cosine Similarity

Prompt Adherence (CLIP Score)	Checks semantic similarity between prompt and image	CLIP ViT-B/32
Frame Consistency (Optical Flow)	Evaluates temporal smoothness between consecutive frames	OpenCV Farneback Optical Flow
Rendering Speed (s/frame)	Average generation time per frame	Internal Timer Logs
GPU Utilization (%)	Indicates hardware efficiency	NVIDIA System Monitor
User Satisfaction (1–5)	Subjective realism and appeal rating	Feedback from 10 participants

Table 1: Evaluation Metrics

7.3 Quantitative Results

Metric	**Value**	**Interpretation**
Identity Preservation	99.94%	Perfect likeness retained
Prompt Adherence (CLIP Score)	0.83 / 1.0	Strong match between visuals and prompts
Frame Consistency (Optical Flow)	0.029	Smooth transitions between frames
Rendering Speed	1.78 s/frame	Efficient generation performance
GPU Utilization	88% (avg)	Optimal hardware usage
User Satisfaction	4.7 / 5	Highly realistic and visually pleasing output

***Table 2:* Quantitative Results**

7.4 Identity Preservation Analysis

To evaluate facial similarity, embeddings of both real and generated faces were extracted using the ArcFace model.
The cosine similarity between 512-dimensional embeddings was calculated as:

$$\text{Cosine Similarity} = \frac{A \cdot B}{|A||B|}$$

where A and B represent feature vectors of the real and generated faces respectively. The average similarity obtained was 0.9994, corresponding to an Identity Preservation Score of 99.94%.
This confirms that the generative process effectively preserved individual facial features without distortion or identity loss.

The near-perfect cosine similarity demonstrates that the LoRA-enhanced diffusion model accurately maintains subject-specific features, achieving nearly identical facial representation in generated media.

7.5 Other Observations

- The Prompt Adherence score of 0.83 reflects strong consistency between textual inputs and visual results, confirming the generator's ability to interpret themes such as "sunset couple walk" or "traditional ceremony shot."

- The Frame Consistency index of 0.029 shows stable transitions between successive frames, ensuring cinematic smoothness and temporal coherence.

- The measured rendering speed of 1.78 seconds per frame and average GPU utilization of 88% verify optimized hardware execution and computational efficiency.

- User evaluation from 10 participants yielded a mean satisfaction rating of 4.7 / 5, indicating highly positive feedback regarding realism, expression fidelity, and overall aesthetics.

Chapter 8

Conclusion & Future Scope

8.1. Conclusion

This project successfully demonstrates the design and implementation of an innovative AI-based system for generating personalized pre-wedding videos. By leveraging state-of-the-art generative models, including Flux.1 and WAN 2.2, within a robust LoRA-based training framework, the system is capable of transforming user-provided images and textual prompts into cinematic video sequences. The methodology, built on a high-performance NVIDIA DGX-v100 platform, establishes a comprehensive pipeline for subject adaptation, prompt-driven image synthesis, and temporally consistent video generation.

The system empowers users to creatively direct their own pre-wedding narrative by specifying backgrounds, attire, and emotions, thereby democratizing access to high-quality, customized video content that has traditionally been costly and time-consuming to produce. The evaluation framework, which combines both qualitative user feedback and quantitative metrics such as identity preservation and prompt adherence, confirms the system's effectiveness in producing visually compelling and emotionally resonant outputs.

While the current implementation provides a strong proof-of-concept, several avenues for future work have been identified to be completed during the 6th semester. These include the integration of advanced image captioning for enhanced semantic understanding, the development of a unified end-to-end user application, and a focused effort on improving facial consistency and model accuracy.

In conclusion, this research not only contributes a novel solution to the field of creative content generation but also lays the groundwork for a more accessible and personalized future in the wedding videography industry. The successful completion of the future work will further evolve this project into a polished, user-centric application, solidifying the potential of AI to revolutionize personal storytelling and memory preservation.

8.2. Future Works

While the current system demonstrates a robust framework for generating personalized pre-wedding videos, several key areas have been identified for future enhancement. The following outlines the planned work to be completed during the **6th semester**, which is aimed at improving the system's accuracy, usability, and creative capabilities.

- **Enhanced Semantic Understanding with Advanced Image Captioning:** The current methodology will be upgraded by integrating a sophisticated, attention-based image captioning model. This will enable the system to automatically generate rich, contextually aware descriptions for uploaded images, capturing nuanced details about subjects and their environment. This advancement is expected to improve the LoRA training process by providing a more accurate textual representation of the visual data, leading to superior prompt adherence.
- **Unified End-to-End System Integration:** A primary objective for the 6th semester is to integrate the distinct pipeline stages (LoRA training, image generation, video synthesis) into a single, cohesive application. This will involve developing a unified backend API to orchestrate the entire workflow, creating a seamless and automated user experience from image upload to final video rendering.
- **Development of an Intuitive User Interface (UI):** To enhance usability, a user-friendly graphical interface will be developed. This UI will provide users with a more intuitive way to interact with the system and will include:
    - **Template-Based Options:** Pre-defined templates for various wedding and pre-wedding themes (e.g., "beach romance," "royal palace," "city nightlife").
    - **Combined Photoshoot and Video Generation:** A feature that allows users to generate both a static photoshoot gallery and a dynamic video from a single set of inputs.
    - **Interactive Previews:** Real-time, low-fidelity previews of the generated content to allow for iterative prompt refinement.

- **Improving Model Accuracy and Face Consistency:** A significant focus will be placed on enhancing the realism and consistency of the generated videos. This will be achieved through several technical improvements to be implemented in the **6th semester**:
    - **Advanced Identity Preservation:** Exploring the use of dedicated facial identity encoders or adopting more robust LoRA training strategies to ensure consistent facial features across different scenes and expressions.
    - **Temporal Consistency Models:** Investigating the use of advanced video models with temporal attention mechanisms to ensure smoother motion and reduce artifacts between frames.
    - **Domain-Specific Fine-Tuning:** Further fine-tuning the models on a larger, more diverse dataset of wedding-themed images and videos to improve their understanding of domain-specific aesthetics.

These planned enhancements aim to evolve the project from its current proof-of-concept stage into a polished, user-centric application, setting a new standard for AI-driven creative content generation in the wedding industry.

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